

Laura Greenstreet



I am a PhD student at Cornell developing AI and optimization methods with applications in biology, including fishery sciences, ecology, and genomics. Previously, I worked with Dr. Claire Kremen to scale an analysis of functional connectivity to the global level and with Dr. Geoffrey Schiebinger to develop optimization methods for single-cell genomics.

EDUCATION

- 2021 - Present **Ph.D. Computer Science**, Cornell University
Research areas: Optimization, Deep Learning, AI/ML for Science
- 2020 **B.Sc. Honours Computer Science, Mathematics Minor**,
University of British Columbia

PUBLICATIONS

-  2025 F.S. Pacheco, S.A. Heilpern, C. DiLeo, ..., **L. Greenstreet**, et al. *Towards sustainable aquaculture in the Amazon*. Nature Sustainability, 2025. <https://doi.org/10.1038/s41893-024-01500-w>
-  2025 Li, H., P. Cote, M. Kuoch, J. Ezike, ..., **L. Greenstreet**, et al. *The dynamics of hematopoiesis over the human lifespan*. Nature Methods, 2025. <https://doi.org/10.1038/s41592-024-02495-0>
-  2025 Massri, A.J., A. Berrio, A. Afanassiev, **L. Greenstreet**, et al. *Single-Cell Transcriptomics Reveals Evolutionary Reconfiguration of Embryonic Cell Fate Specification in the Sea Urchin *Heliocidaris erythrogramma**. Genome Biology and Evolution, 2025. <https://doi.org/10.1093/gbe/evae258>
-  2023 **Greenstreet, L.**, A. Afanassiev, Y. Kijima, M. Heitz, S. Ichiguro, et al. *DNA-GPS: A Theoretical Framework for Optics-Free Spatial Genomics and Synthesis of Current Methods*. Cell Systems, 2023. <https://doi.org/10.1016/j.cels.2023.08.005>
-  2023 T.M. Nolan, N. Vukasinovic, C.W. Hsu, J. Zhang, ..., **Laura Greenstreet**, et al. *Brassinosteroid gene regulatory networks at cellular resolution in the Arabidopsis root*. Science, 2023. <https://doi.org/10.1126/science.adf4721>
-  2023 Mirka, R., **L. Greenstreet**, M. Grimson, C.P. Gomes. *A New Approach to Finding $2 \times n$ Partially Spatially Balanced Latin Rectangles*, CP, 2023.
-  2022 Brennan, A., R. Naidoo, **L. Greenstreet**, Z. Mehrabi, N. Ramankutty, C. Kremen. *Functional Connectivity of the Worlds Protected Areas*. Science, 2022. <https://doi.org/10.1126/science.abl8974>
-  2022 **Greenstreet, L.**, N.J.A. Harvey, V. Sanches Portella. *Efficient and Optimal Fixed-Time Regret with Two Experts*. ALT, 2022. <https://doi.org/10.48550/arXiv.2203.07577>
-  2021 Zhang, S., A. Afanassiev, **L. Greenstreet**, T. Matsumoto, G. Schiebinger. *Optimal transport analysis reveals trajectories in steady-state systems*. PLOS Computational Biology, 2021. <https://doi.org/10.1371/journal.pcbi.1009466>
-  2021 Li, H., J. Ezike, A. Afanassiev, **L. Greenstreet**, et al. *Single Cell Analysis Elucidates the Maturation of Human Stem and Progenitor Cell Function from Fetal through Adult Hematopoiesis*. Blood, 2021. <https://doi.org/10.1182/blood-2021-151090>
-  2021 Shahan R., C. Hsu, T.M. Nolan, B.J. Cole, I.W. Taylor, **L. Greenstreet**, et al. *A single cell Arabidopsis root atlas reveals developmental trajectories in wild type and cell identity mutants*. Developmental Cell, 2021. <https://doi.org/10.1016/j.devcel.2022.01.008>

 Bioinformatics  Geospatial  Theory

- 2020 Massri, A.J., **L. Greenstreet**, A. Afanassiev, A. Berrio Escobar, G.M. Wray, G. Schiebinger, D.R. McClay. *Developmental Single-cell transcriptomics in the Lytechinus variegatus Sea Urchin Embryo*. Development, 2020. <https://doi.org/10.1242/dev.198614>

WORKSHOPS AND PRESENTATIONS

- 2023 **Greenstreet, L**, F. Pacheco, J. Fan, M. Eichemberger Ummus, et al. *Detecting Aquaculture with Deep Learning in a Low-Data Setting*. AGU Fall Meeting Abstracts.
- 2023 **Greenstreet, L**, J. Fan, F. Siqueira Pacheco, Y. Bai, M. Eichemberger Ummus, et al. *Detecting Aquaculture with Deep Learning in a Low-Data Setting*. SigKDD 2023 Fragile Earth Workshop.
- 2020 **Greenstreet, L**, and E. Lai. *Developing a Data-Driven Electric Vehicle Strategy in Surrey, BC*. SigKDD 2020 Social Impact Session.

AWARDS

- 2022-2024 **Graduate Teaching Award x5**, Department of Computer Science, Cornell University
- 2020 **Undergraduate Summer Research Award**, Natural Sciences and Engineering Research Council of Canada (NSERC)
- 2019 **Data Science for Social Good Fellowship**, UBC Data Science Institute
- 2018 **Stanley M Grant Scholarship in Mathematics**, UBC Department of Mathematics

TECHNICAL SKILLS

3+ Years Experience: Python, Git, Linux

1-3 Years Experience: R, Julia, Matlab, Java, SQL

Libraries/Tools: Pytorch, QGIS, Postgres, Gurobi

Experience: deep learning: CNNs, GNNs, VAEs, transformers, contrastive learning, manifold learning, mathematical programming, SQL

RESEARCH EXPERIENCE

- 05/2024 - 08/2024 **Graduate Research Assistant**, Cornell University
- 05/2023 - 01/2024 **Graduate Research Assistant**, Computational Sustainability Lab, Department of Computer Science, Cornell University
- 05/2022 - 08/2022 **Research Assistant**, Schiebinger Lab, Department of Mathematics, University of British Columbia
- 09/2019 - 09/2020 **Research Assistant**, WoRCS Lab, Institute for Resources, Environment, and Sustainability, University of British Columbia
- 05/2019 - 08/2019 **Fellow - Data Science for Social Good Program**, University of British Columbia Data Science Institute

TEACHING EXPERIENCE

FA23, SP23, SP24	Head Teaching Assistant , Cornell University, Ithaca, NY CS 2700 - Excursions in Computational Sustainability CS 4700/4701 - Foundations/Practicum in Artificial Intelligence
FA22, SP22, SP25	Teaching Assistant , Cornell University, Ithaca, NY CS 4820 - Introduction to Analysis of Algorithms CS 4220 - Numerical Analysis: Linear and Nonlinear Problems CS 3220 - Computational Mathematics for Computer Science

COMMUNITY INVOLVEMENT

2024	Organizer , AI for Science Seminar
2023	Organizer , NeurIPS Computational Sustainability Workshop
2023	Mentor , BURE Undergraduate Research Program, Cornell
2023-2024	Assistant Organizer , AI for Science Program, Cornell
2020-2021	Mentor , Data Science for Social Good Program, UBC Data Science Institute